

Abhi Patel

apatel7@wpi.edu | (401)744-2940 | U.S. Citizen | [linkedin.com/in/abhipatel017](https://www.linkedin.com/in/abhipatel017) | Portfolio: abhi5301.github.io

EDUCATION

Worcester Polytechnic Institute (WPI)

Worcester, MA | May 2028

B.S. Robotics Engineering, M.S. Robotics Engineering

GPA 3.83

Relevant Coursework: Metal Additive, Industrial Robotics, Robotic Manipulation, Embedded Systems, Control Engineering

SKILLS

CAD & Design: SolidWorks (CSWP), Onshape, Fusion 360, Ansys, KiCad

Manufacturing & Fabrication: FDM/SLA Additive Manufacturing, CNC Machining, Laser Cutting, PCB Design

Programming & Tools: PLC Structured Text, ROS 2, Python, C++, Java, MATLAB, Git/GitHub, VSCode, Ubuntu/Linux

MECHANICAL DESIGN & MANUFACTURING

Mechanical Design & Fabrication Lead | Secretary, Vex U Robotics Competition

Aug 2024 - Present

- Led mechanical design and fabrication for a robotics team, mentoring members and elevating team wide skill
- Designed a dual-robot ascent system with manufacturability in mind, reducing climb time to five seconds
- Applied parametric CAD modeling, pneumatics, and mixed-material design to meet subsystem requirements
- Manufactured integrated components using 3D printing, machining, and laser cutting, streamlining production

Differential Swerve Drive Module

Jun 2025 - Aug 2025

- Redesigned the drivetrain into a swerve system, enabling full holonomic motion and doubling system traction
- Designed a compact differential gear mechanism to deliver full motor power while maintaining total orientation control
- Engineered wear-resistant bearings by inlaying steel wire into FDM printed parts to improve durability under motion
- Redesigned components for additive manufacturing, applying variable layer heights to solve geometry challenges

Multimaterial FDM 3D Printer

May 2026 - Present

- Assembled and built an open source 3D printer design, including the full motion system, tool head, and firmware
- Configured interchangeable tool heads to enable printing different thermoplastics and materials within a single part
- Built a custom web interface and macros to remotely manage prints and optimize print calibration and startup

Additive Manufacturing Lab Technician, Worcester, MA

Sep 2025 - May 2026

- Troubleshoot and maintained a 3D print farm with 8+ FDM and resin printers, maintaining a high uptime percentage
- Optimized print parameters for multi-material jobs (ABS, TPU, Nylon), improving print success rates and part lifespan

Mechanisms & Airbrakes, High Power Rocketry Club

Aug 2024 - May 2025

- Redesigned rocket airbrakes with a 30% smaller footprint, collaborating with avionics and propulsion subteams

ROBOTICS & CONTROLS

Industrial Robotic Automation System

Mar 2026 - May 2026

- Integrated two UR5e robotic arms and a Fanuc Delta manipulator, all 6-DOF, into a coordinated multi-robot sorting cell
- Programmed an Arduino-based PLC in structured text to orchestrate high-level control across all three robot systems
- Utilized vision cameras to detect and localize randomized parts for automated pick-and-place sorting
- Synchronized conveyors and sensors to palletize sorted parts into a structured grid layout

SLAM Pathing and Navigation System

Mar 2026 - May 2026

- Developed a ROS2 based SLAM system in Python to localize and map a robot in real time using lidar sensor data
- Designed a path optimization algorithm to minimize traversal time between arbitrary points on the generated map
- Built exception handling and edge case testing to ensure reliable performance under sensor uncertainty
- Executed rigorous empirical field trials across diverse operating conditions to ensure robust overall functionality

Embedded Sensor Interface PCB

Dec 2025 - May 2026

- Designed a pcb to interface embedded I2C sensors with a central robot control system for odometry and localization
- Utilized onboard power regulation system, RP2040 processor, and RS485 serial communication to centralized system
- Implemented custom firmware in C++ to manage I2C and RS485 communication protocols and odometry calculations

Advanced Robotics, Teaching Assistant, Worcester, MA

Jan 2026 - Present

- Guided students in advanced robotics courses on embedded software, sensor integration, and navigation/pathing